Aytekin Erdogan

(+90) 537 561 7103
aytekerdo@gmail.com
aytekerdo
aytekXR

Experience

Nov 2025 – Present **Technical Strategy Engineer**, BEYOND TECHNOLOGIES.
- Shaping the company's product and technology roadmap and steering new R&D initiatives through in-depth market intelligence and competitive analysis in the mini-UAV domain
- Representing the company at domestic and international engagements, including high-level meetings and industry events, directly interfacing with C-level stakeholders
- Providing technical expertise during the sales process, including client meetings and negotiations
- Representing the company at international fairs and industry events
- Hosting VIP guests, delegations, and project-related visitors on behalf of the company

Apr 2025 – Nov 2025 **Tech Lead, AI & Autonomy (TARNAV)**, BEYOND TECHNOLOGIES.
- Led a multi-national Europe-based,the TARNAV, autonomy kit development project as tech lead within the AI team
- Designed system architecture and guided implementation for the autonomous kit
- Managed cross-functional development with a focus on scalable, high-quality systems

Apr 2024– Apr 2025 **Tech Lead, AI Product**, EXIN HEALTH.
- Leading development of the ORGenie product to improve operating room intelligence
- Developing and deploying AI models for surgical data analysis
- Specializing in instance segmentation, stereo vision, and automatic speech recognition
- Implementing data privacy solutions compliant with GDPR and HIPAA
- Creating a video event annotation tool for medical procedures

Jun 2022– Apr 2025 **Software Engineer- ML Research**, ERICSSON.
- Recognized as Key Contributor in 2024 and awarded stock options
- Developed enterprise-level software for 5G/6G Transport Networks (TN) with AI/ML integration
- Senior C++ Developer for 5G/6G technologies

Sep 2021– May 2022 **Software Engineer, Autonomous Driving**, FORD OTOSAN.
- Designed and implemented a multi-sensor calibration and alignment framework
- Conducted research on monocular 2D/3D object detection for autonomous vehicles
- Developed perception systems for obstacle detection in autonomous driving

Aug 2018– Apr 2021 **Software Engineer, Computer Vision**, METEKSAN DEFENCE.
- Created Board Support Packages for Linux distributions
- Developed C++ applications for Xilinx UltraScale Architecture
- Implemented face recognition and license plate recognition systems
- Deployed computer vision solutions on NVIDIA Jetson platforms

Education

PhD 2028 **Middle East Technical University**.
- Graduate School of Informatics/Computer Science
- Research focus: Computer Vision and Deep Learning

MSc 2023 **Middle East Technical University**.
- Graduate School of Informatics/Computer Science
- Thesis: A Transformer-Based Approach for Fusing Infrared and Visible Band Images

BSc 2019 **Hacettepe University**.
- Electrical and Electronics Engineering
- Rank: 8/130, GPA: 3.3/4.00

HS 2015 **Isiklar Air Force Military High School**, ***Rank***: *11/171* ***GPA***: *80.03/100*.

Technical Skills

Languages	C/C++, Python, MATLAB
Technologies	C++17, OpenMP, CUDA C/C++, PyTorch, OpenCV, Boost
Focus	Computer Vision, Machine Learning, Deep Learning, Embedded Systems

Publications

Aytekin Erdogan Erdem Akagunduz. FuseFormer: A Transformer for Visual and Thermal Image Fusion. *arXiv.org pre-print*, 2024.

Aytekin Erdogan. A Transformer-Based Approach for Fusing Infrared and Visible Band Images. Master's thesis, Middle East Technical University, 2023.

Aytekin Erdogan. 3D-printed Reflectarray Antenna Design and Fabrication. *Bachelor of Science degree thesis, Hacettepe University*, 2019.

Aytekin Erdogan. Total Quality Management in Training of Airmen. *IJAS Conference, Nevada USA*, 2016.

Projects

2023 **CUDA Performance Analysis for Computer Vision**, High Performance Computing 🏠.
Implemented a computer vision pipeline from scratch comparing single-thread CPU, OpenMP, and CUDA performance. Conducted detailed timing analysis. Project Link

2022 **Lane Detection: A Comparative Study**, Computer Vision 🏠.
Developed and compared various lane detection algorithms using traditional computer vision techniques. Project Link

2021 **Low-Light Image Enhancement**, Image Processing 🏠.
Developed solutions for enhancing images captured in low-light conditions using advanced image processing techniques.

2021 **Real-time Multi-Person Pose Estimation**, Deep Learning 🏠.
Implemented and optimized a lightweight version of OpenPose for real-time human pose detection.

2021 **Real-Time Semantic Segmentation**, Deep Learning 🏠.
Developed a semantic segmentation network for autonomous vehicles with real-time performance.

2019 **Expiration Date Detection System**, Image Processing 🏠.
Built an FPGA-based system to detect and verify food expiration dates using computer vision.

2019 **IoT-Based Home Automation System**, Internet of Things 🏠.
Designed and implemented a cost-effective smart home system with remote control capabilities.

Certifications

May 2023 **C and C++ SEI CERT Based Secure Coding**, by Scademy, 🏠.
Mar 2022 **Self Driving Car Engineer**, by Udacity, 🏠.
Jan 2022 **Scrum Team Member**, by Scrum Inc, 🏠.
Jun 2020 **TensorFlow Developer Specialization**, by deeplearning.ai, 🏠.
Apr 2020 **Deep Learning Specialization**, by deeplearning.ai, 🏠.
Jan 2020 **Machine Learning**, by Stanford University, 🏠.

Additional Information

Languages	**English (IELTS 7.5/9, TOEFL 102/120), Turkish (Native)**.
Interests	**Social Entrepreneurship, Technology for Social Good**.
Activities	**Debate (Former National Team Candidate), Swimming (Former Professional)**.